Vaccine tender scheduling: single-level and bilevel formulations

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This document states the current single-level UG tender scheduling model (Uhorchak et al.) and its planned bilevel extension, in which UG is the upper level and the producer profit model MP is the lower level. Equation numbers continue across both.

1. Nomenclature

Symbols marked † are introduced for the bilevel model; all others follow the manuscript.

Sets

Symbol	Description
$T = \{1, \ldots, N\}$	Planning periods (years), horizon length N
A	Antigens
V	Vaccines
P	Pre-qualified producers
M	All-unit discount brackets
$\mathcal{W} = \{(t, \tau) : t \leqslant \tau \leqslant \min(t + \Delta - 1, N)\}$	Feasible tender windows
$A_p \subseteq A$, $V_p \subseteq V$	Antigens and vaccines producer p supplies
P_a, $P_v \subseteq P$	Producers able to supply antigen a, vaccine v
$A_v \subseteq A$, $V_a \subseteq V$	Antigens in vaccine v; vaccines containing antigen a
K †	Price ladder levels
J †	Award-share levels
$\mathcal{W}(l) = \{(t, \tau) \in \mathcal{W} : t \leqslant l \leqslant \tau\}$ †	Tender windows active in period l

Parameters

Symbol	Description
δ_t	Time-discount factor for period t
g_t	Tender set-up cost
r̄_vt	Reference (reservation proxy) price per dose of vaccine v
r_vpt	Reservation price of producer p for vaccine v
ζ_vm	Discount rate of bracket m, increasing in m
ϕ^L_vm, ϕ^U_vm	Lower and upper quantity bounds of bracket m
β	Penalty per missed dose
h_v	Holding cost as a fraction of price
d_at	Demand for antigen a in period t
Î_v0	Initial inventory
s̄_p	Base capacity of producer p, shared across V_p
s_pt	Available capacity of producer p in period t
κ	Allowable capacity expansion rate per unit of L
γ_p	Cost per unit of capacity expansion
f_vptτ	Production set-up cost for a commitment over (t, τ)
l_vp	Required return margin in the ROI constraint
Δ	Maximum tender length
λ_k †	Price multiplier of ladder level k
ρ_vtk = λ_k r̄_vt †	Price per dose at level k
υ_j †	Award share of level j, in [0, 1]
d̂_vt †	Vaccine-level demand forecast
D̂_vtτ = Σ_{l=t}^{τ} d̂_vl †	Tender volume of vaccine v over (t, τ)
c_vp †	Per-dose production cost (synthetic)

Symbol	Description
L̄_p †	Upper bound on capacity expansion units

Decision variables

Symbol	Domain	Description	Bilevel owner
F_atτ	{0, 1}	Tender for antigen a opened over (t, τ)	Leader
Π_vtk †	{0, 1}	Price level k set for vaccine v in tenders starting at t	Leader
Ω_vptτj †	{0, 1}	Award-share level j for producer p, vaccine v, window (t, τ)	Leader
I_vt	⩾ 0	Inventory of vaccine v at end of t	Leader
V_vt	⩾ 0	Doses of vaccine v administered in t	Leader
S_at	⩾ 0	Cumulative missed doses of antigen a	Leader
W_ptτ	{0, 1}	Producer p commits to window (t, τ)	Follower p
Y_pt	{0, 1}	Producer p produces in t	Follower p
Z_vptm	{0, 1}	Bracket m active for (v, p, t)	Follower p
L_pt	ℤ+	Capacity expansion units	Follower p
Q_vptτm	⩾ 0	Committed quantity in bracket m	Follower p
X_vpt	⩾ 0	Doses delivered in t (single-level model)	Follower p
X_vptτl †	⩾ 0	Doses delivered in l against window (t, τ)	Follower p
Γ_vptτmk †	⩾ 0	Part of Q_vptτm priced at level k	Follower p

2. Single-level model (current)

In the single-level model a coordinating entity chooses every variable. The UNICEF-Gavi objective (UG) minimizes discounted tendering, procurement, missed-dose and holding costs. The producer profit objective (MP) is stated after it because it becomes the lower

level of the bilevel model.

Model UG

$$\min\ \sum_{t\in T}\delta_t\Biggl(\sum_{a\in A}\sum_{\tau:(t,\tau)\in\mathcal F} g_t F_{at\tau}
+\sum_{v\in V}\sum_{p\in P_v}\sum_{m\in M}\sum_{\tau:(t,\tau)\in\mathcal F}\bar r_{vt}(1-\zeta_{vm})\,Q_{vpt\tau m}
+\sum_{a\in A}\beta S_{at}+\sum_{v\in V}h_v\bar r_{vt}I_{vt}\Biggr) \qquad (1)$$

subject to

$$
\begin{aligned}
&(\tau-t+1)\,F_{at\tau}\le \sum_{p\in P_a}\sum_{l=t}^{\tau}Y_{pl} && \forall a\in A,\ (t,\tau)\in\mathcal F && (2)\\
&\sum_{\tau:(t,\tau)\in\mathcal F}F_{at\tau}\le 1 && \forall a\in A,\ t\in T && (3)\\
&\sum_{(l,k)\in\mathcal F:\ l\le t\le k}F_{alk}\ge 1 && \forall a\in A,\ t\in T && (4)\\
&\sum_{a\in A_p}F_{at\tau}\ge W_{pt\tau} && \forall p\in P,\ (t,\tau)\in\mathcal F && (5)\\
&F_{at\tau}\le \sum_{p\in P_a}W_{pt\tau} && \forall a\in A,\ (t,\tau)\in\mathcal F && (6)\\
&\sum_{v\in V_p}\sum_{m\in M}Q_{vpt\tau m}\le W_{pt\tau}\Bigl(\sum_{l=t}^{\tau}\bar s_p+\sum_{k=1}^{t}\kappa(\tau-t+1)\,\bar s_p L_{pk}\Bigr) && \forall p\in P,\ (t,\tau)\in\mathcal F && (7)\\
&\sum_{v\in V_p}\sum_{m\in M}\bar r_{vt}(1-\zeta_{vm})Q_{vpt\tau m}\ge \sum_{v\in V_p}(1+l_{vp})f_{vpt\tau}W_{pt\tau} && \forall p\in P,\ (t,\tau)\in\mathcal F && (8)\\
&\phi^L_{vm}Z_{vptm}\le \sum_{\tau:(t,\tau)\in\mathcal F}Q_{vpt\tau m} && \forall v,\ p\in P_v,\ t,\ m && (9)\\
&\phi^U_{vm}Z_{vptm}\ge \sum_{\tau:(t,\tau)\in\mathcal F}Q_{vpt\tau m} && \forall v,\ p\in P_v,\ t,\ m && (10)\\
&\sum_{m\in M}Z_{vptm}=1 && \forall v,\ p\in P_v,\ t && (11)\\
&\sum_{m\in M}Q_{vpt\tau m}\ge \sum_{l=t}^{\tau}X_{vpl} && \forall v,\ p\in P_v,\ (t,\tau)\in\mathcal F && (12)\\
&\sum_{v\in V_p}X_{vpt}\le Y_{pt}\Bigl(s_{pt}+\sum_{l=1}^{t}\kappa\bar s_pL_{pl}\Bigr) && \forall p\in P,\ t\in T && (13)\\
&I_{v,t-1}+\sum_{p\in P_v}X_{vpt}=V_{vt}+I_{vt} && \forall v\in V,\ t\in T && (14)\\
&d_{at}-\sum_{v\in V_a}V_{vt}+S_{a,t-1}\le S_{at} && \forall a\in A,\ t\in T && (15)\\
&I_{v0}=\hat I_{v0} && \forall v\in V && (16)\\
&F_{at\tau},Y_{pt},W_{pt\tau},Z_{vptm}\in\{0,1\},\quad L_{pt}\in\mathbb Z_+ && && (17)\\
&Q_{vpt\tau m},X_{vpt},I_{vt},S_{at},V_{vt}\ge 0 && && (18)
\end{aligned}
$$

The manuscript's social-benefit model (SB), equation (19), is outside the scope of this document.

Model MP (lower level of the bilevel model) maximizes producer net revenue subject to (2)–(18), written as a minimization:

$$\min\ \sum_{t\in T}\delta_t\Biggl(-\sum_{v\in V}\sum_{p\in P_v}\sum_{m\in M}\sum_{\tau:(t,\tau)\in\mathcal F}\bar r_{vt}(1-\zeta_{vm})\,Q_{vpt\tau m} +\sum_{p\in P}\gamma_pL_{pt}+\sum_{p\in P}\sum_{v\in V_p}\sum_{\tau:(t,\tau)\in\mathcal F}f_{vpt\tau}W_{pt\tau}\Biggr) \qquad (20)$$

Transcription notes: the PDF does not show the upper index of the inner capacity-expansion sum in (7); it is written here as k ⩽ t (expansions made by the tender start). Sums over τ in (9)–(10) are written over feasible windows starting at t.

3. Linearization

Constraints (7) and (13) multiply a binary by the integer L. With $L_{pt} \leqslant \bar L_p$, each product is replaced exactly by a continuous auxiliary variable and four McCormick inequalities (McCormick, 1976; Glover, 1975).

For (7), let $\omega_{pt\tau k} = W_{pt\tau} L_{pk}$:

$$\begin{aligned}
&\omega_{pt\tau k}\le \bar L_p W_{pt\tau},\quad \omega_{pt\tau k}\le L_{pk},\quad \omega_{pt\tau k}\ge L_{pk}-\bar L_p(1-W_{pt\tau}),\quad \omega_{pt\tau k}\ge 0 && \forall p,\ (t,\tau)\in\mathcal F,\ k\le t && (21)\\
&\sum_{v\in V_p}\sum_{m\in M}Q_{vpt\tau m}\le (\tau-t+1)\,\bar s_p W_{pt\tau}+\kappa(\tau-t+1)\,\bar s_p\sum_{k=1}^{t}\omega_{pt\tau k} && \forall p,\ (t,\tau)\in\mathcal F && (22)
\end{aligned}$$

For (13), let $\eta_{ptl} = Y_{pt} L_{pl}$:

$$\begin{aligned}
&\eta_{ptl}\le \bar L_p Y_{pt},\quad \eta_{ptl}\le L_{pl},\quad \eta_{ptl}\ge L_{pl}-\bar L_p(1-Y_{pt}),\quad \eta_{ptl}\ge 0 && \forall p,\ t,\ l\le t && (23)\\
&\sum_{v\in V_p}X_{vpt}\le s_{pt}Y_{pt}+\kappa\bar s_p\sum_{l=1}^{t}\eta_{ptl} && \forall p,\ t && (24)
\end{aligned}$$

(22) replaces (7) and (24) replaces (13) in every model below. Both products involve only producer variables, so in the bilevel model the linearization sits entirely inside each follower.

4. Bilevel model (planned)

The bilevel model places UG over MP. The coordinating entity (leader, objective (1)) announces the tender calendar, per-dose prices and award shares. Each producer (follower, objective (20) restricted to its own variables) then chooses its participation, commitments, capacity and deliveries to maximize its own profit. Let x = (F, Π, Ω) denote the linking decisions and y_p = (W, Y, Z, L, Q, X, Γ, ω, η) the variables of producer p.

Assumptions

1. Open-loop commitment: the leader fixes the full calendar and terms at the start of the horizon, consistent with published multi-year tender calendars.
2. Producers do not cooperate and share no constraint; they interact only through the leader's decisions.
3. Complete information: the leader knows each producer's costs and capacities (synthetic in the experiments).
4. Optimistic convention: among a producer's optimal responses, the one best for the leader is selected.
5. Prices are chosen from a finite ladder, and discount brackets (ζ, ϕ) are announced in advance as a menu from which producers self-select.

Upper level (leader)

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\min_{x,\,I,\,V,\,S}\ \sum_{t\in T}\delta_t\Biggl(\sum_{a\in A}\sum_{\tau:
(t,\tau)\in\mathcal F} g_t F_{at\tau}
+\sum_{v\in V}\sum_{p\in P_v}\sum_{\tau:(t,\tau)\in\mathcal F}\sum_{m\in
M}\sum_{k\in K}\rho_{vtk}(1-\zeta_{vm})\,\Gamma_{vpt\tau mk}
+\sum_{a\in A}\beta S_{at}+\sum_{v\in V}h_v\bar r_{vt}I_{vt}\Biggr) \qquad
(25)
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subject to (3), (4), (15), (16) and

$$
\begin{aligned}
&\sum_{k\in K}\Pi_{vtk}\le \sum_{a\in A_v}\sum_{\tau:(t,\tau)\in\mathcal F}F_{at\tau} && \forall v\in V,\ t\in T && (26)\\
&\sum_{k\in K}\Pi_{vtk}\le 1 && \forall v\in V,\ t\in T && (27)\\
&\sum_{j\in J}\Omega_{vpt\tau j}\le \sum_{a\in A_v}F_{at\tau} && \forall v\in V,\ p\in P_v,\ (t,\tau)\in\mathcal F && (28)\\
&\sum_{j\in J}\Omega_{vpt\tau j}\le 1 && \forall v\in V,\ p\in P_v,\ (t,\tau)\in\mathcal F && (29)\\
&I_{v,t-1}+\sum_{p\in P_v}\sum_{(t',\tau)\in\mathcal F(t)}X_{vpt'\tau t}=V_{vt}+I_{vt} && \forall v\in V,\ t\in T && (30)\\
&F_{at\tau},\ \Pi_{vtk},\ \Omega_{vpt\tau j}\in\{0,1\};\quad I_{vt},V_{vt},S_{at}\ge 0 && && (31)\\
&y_p\in\mathcal R_p(x) && \forall p\in P && (32)
\end{aligned}
$$

Lower level (follower p)

$$
\mathcal R_p(x)=\arg\max_{y_p}\ \sum_{t\in T}\delta_t\Biggl(\sum_{v\in V_p}\sum_{\tau:(t,\tau)\in\mathcal F}\sum_{m\in M}\sum_{k\in K}\rho_{vtk}(1-\zeta_{vm})\,\Gamma_{vpt\tau mk}
-\gamma_{pL_{pt}}-\sum_{v\in V_p}\sum_{\tau:(t,\tau)\in\mathcal F}f_{vpt\tau}W_{pt\tau}
-\sum_{v\in V_p}\sum_{(t',\tau)\in\mathcal F(t)}c_{vp}X_{vpt'\tau t}\Biggr)
\qquad (33)
$$

Objective (33) is the MP objective (20) for producer p alone, with revenue priced at the leader's chosen level and one added term, the production cost c_vp. Setting c_vp = 0 recovers (20) exactly. It is subject to (21)–(23), (9)–(11) for v ∈ V_p, and

$$
\begin{aligned}
&W_{pt\tau}\le \sum_{a\in A_p}F_{at\tau} && \forall (t,\tau)\in\mathcal F && (34)\\
&\sum_{m\in M}Q_{vpt\tau m}\le \hat D_{vt\tau}\sum_{j\in J}\upsilon_j\,\Omega_{vpt\tau j} && \forall v\in V_p,\ (t,\tau)\in\mathcal F && (35)\\
&\sum_{m\in M}Q_{vpt\tau m}\le \hat D_{vt\tau}\,W_{pt\tau} && \forall v\in V_p,\ (t,\tau)\in\mathcal F && (36)\\
&\sum_{k\in K}\Gamma_{vpt\tau mk}=Q_{vpt\tau m} && \forall v\in V_p,\ (t,\tau)\in\mathcal F,\ m && (37)\\
&\Gamma_{vpt\tau mk}\le \hat D_{vt\tau}\,\Pi_{vtk} && \forall v\in V_p,\ (t,\tau)\in\mathcal F,\ m,\ k && (38)\\
&\sum_{l=t}^{\tau}X_{vpt\tau l}=\sum_{m\in M}Q_{vpt\tau m} && \forall v\in V_p,\ (t,\tau)\in\mathcal F && (39)\\
&\sum_{v\in V_p}\sum_{(t',\tau)\in\mathcal F(t)}X_{vpt'\tau t}\le s_{pt}Y_{pt}+\kappa\bar s_p\sum_{l=1}^{t}\eta_{ptl} && \forall t\in T && (40)\\
&W_{pt\tau},Y_{pt},Z_{vptm}\in\{0,1\};\ L_{pt}\in\{0,\dots,\bar L_p\};\ Q,X,\Gamma,\omega,\eta\ge 0 && && (41)
\end{aligned}
$$

Remarks. (37)–(38) price each commitment exactly at the chosen ladder level without a bilinear term, so every follower has constant objective coefficients and the leader's variables enter only the right-hand sides, the setting assumed by exact MIBLP methods. Setting $y_p = 0$ is feasible for every x and yields zero profit, so individual rationality holds without the ROI constraint (8). Constraints (30) and (15) contain follower deliveries but are always satisfiable through V, I and S, so the only coupling between levels is through the leader's objective.

5. Equivalent forms and bounds

Single-follower equivalent. Let $\mathcal{R}_p(x)$ be the feasible set of follower p, defined by (9)–(11), (21)–(23) and (34)–(41), and $\pi_p(x, y_p)$ its objective (33). The sets $\mathcal{R}_p(x)$ share no constraint and the objectives are separable, so

$$
\prod_{p\in P}\mathcal R_p(x)=\arg\max\Bigl\{\sum_{p\in P}\pi_p(x,y_p)\ :\ y_p\in\mathcal Y_p(x)\ \ \forall\ p\in P\Bigr\} \qquad (42)
$$

The model can therefore be passed to a single-follower MIBLP solver such as MibS (Tahernejad, Ralphs & DeNegre, 2020) with the summed producer profit as the lower-level objective, while decomposition methods can still solve each producer's problem separately.

Value-function reformulation. With $\varphi_p(x) = \max\{\pi_p(x, y_p) : y_p \in \mathcal{Y}_p(x)\}$, the bilevel model is equivalent to the single-level problem

$$\min\ (25)\quad\text{s.t.}\quad (3),(4),(15),(16),(26)\text{--}(31),\quad y_p\in\mathcal Y_p(x),\quad \pi_p(x,y_p)\ge \varphi_p(x)\quad \forall\ p\in P \qquad (43)$$

The last constraint is nonconvex and defined implicitly; reformulation-and-decomposition algorithms (Zeng & An, 2014; Yue, Gao, Zeng & You, 2019) approximate φ_p from below by enumerating follower responses and converge finitely when the linking variables x are integer, which holds here.

High-point relaxation. Dropping $\pi_p \geqslant \varphi_p$ from (43) gives the high-point relaxation (HPR), a single MILP:

$$v^{HPR}=\min\ (25)\quad\text{s.t.}\quad (3),(4),(15),(16),(26)\text{--}(31),\quad y_p\in\mathcal Y_p(x)\ \ \forall\ p\in P \qquad (44)$$

For the bilevel optimal value v^{BL}, $v^{HPR} \leqslant v^{BL}$. The cost of producer autonomy is defined as

$$\mathrm{CPA}=\frac{v^{BL}-v^{HPR}}{v^{HPR}} \qquad (45)$$

With one price level ($|K| = 1$, $\lambda_1 = 1$), one award level ($u = 1$), and constraints (2), (6) and (8) restored, the HPR reduces to model UG apart from the tender index on deliveries. UG with (8) is therefore the manuscript's approximation of producer rationality, and its schedule can be evaluated against producers' true best responses.

6. Constraint map

Where each element of the single-level model goes in the bilevel model.

Single-level	Bilevel	Change
Objective (1)	Leader objective (25)	Fixed price $\bar r_{vt}$ replaced by chosen ladder price through Γ

Single-level	Bilevel	Change
Objective (20)	Follower objective (33)	Split per producer; adds production cost c_vp
(2) tender requires production	Removed	Producers' Y is their own choice; an unserved tender appears as missed doses
(3), (4) tender structure	Leader	Unchanged
(5) commit only to open tenders	Follower (34)	F is a parameter to the follower
(6) tender requires a committed producer	Removed	Same reason as (2)
(7) commitment capacity	Follower (21)–(22)	Linearized
(8) ROI	Removed	Replaced by follower optimality; zero response always feasible
(9)–(11) discount brackets	Follower	Producers self-select brackets
(12) commitment covers deliveries	Follower (39)	Equality on tender-indexed deliveries
(13) per-period capacity	Follower (23), (40)	Linearized; deliveries summed over active windows
(14) inventory balance	Leader (30)	Deliveries come from followers
(15), (16) missed doses, initial stock	Leader	Unchanged
New	Leader (26)–(29)	Price ladder and award shares
New	Follower (35)–(38)	Award ceiling, commitment link, price split

References. Glover (1975), Management Science 22(4); McCormick (1976), Mathematical Programming 10; Tahernejad, Ralphs & DeNegre (2020), Mathematical Programming Computation 12; Yue, Gao, Zeng & You (2019), Journal of Global Optimization 73(1); Zeng & An (2014), Optimization Online.